YongKyung Oh, Ph.D.

yongkyung.oh@pusan.ac.kr
yongkyungoh@outlook.com
+82-51-510-6624
+82-10-2671-8334

Assistant Professor at Pusan National University

Data Science Major, School of BioMedical Convergence Engineering

Webpage | Scientific Inference Lab | GitHub | LinkedIn | Google Scholar | ORCID | Semantic Scholar | Web of Science | Scopus | DBLP

Biography

Dr. YongKyung Oh is an Assistant Professor in the Data Science Major, School of BioMedical Convergence Engineering, Pusan National University. His research interests span AI for Science, industrial AI and decision-making, evidence-centered AI, and continuous-time modeling, with an emphasis on scientific discovery and informed decisions. He previously conducted postdoctoral research at the University of California, Los Angeles (UCLA), from 2023 to 2026, and at the Ulsan National Institute of Science and Technology (UNIST). His earlier research experience includes General Motors (GM) and Carnegie Mellon University (CMU). He earned a B.Sc. in Physics, a Master in Technology and Innovation Management, and a Ph.D. in Industrial Engineering, all from UNIST. His recent work includes a sole-authored ICML 2026 position paper on evidence standards. His distinctions include the ICLR 2024 Spotlight, the CHIL 2025 Best Paper Award, and a nomination for the 2025 UCLA Chancellor's Award for Postdoctoral Research. His research support includes the NRF Postdoctoral Fellowship for Overseas Research as Principal Investigator and an NVIDIA Academic Grant as Lead Scientist.

Research Interests

AI for Science: Combining machine learning, scientific knowledge, and simulation to discover mechanisms and understand complex systems. Research directions include inverse problems and system identification, simulation-based surrogate models, and experimental design for scientific discovery.

Industrial AI & Decision-Making: Connecting machine learning, optimization, and simulation to support decisions in manufacturing, logistics, and service systems. Research directions include quality, reliability, and predictive maintenance; resource allocation; and production and operations planning.

Evidence-Centered AI: Developing evaluation principles that align AI claims with robust comparisons, uncertainty assessment, and reproducible evidence. This direction addresses model validity, distribution shift, and failures that aggregate performance measures can conceal.

Continuous-Time Modeling: Learning continuous dynamics from irregular observations through neural differential equations, with attention to stability and uncertainty. The aim is to connect dynamical representations with prediction, system identification, and simulation.

Professional Experience

School of BioMedical Convergence Engineering, Pusan National University Current appointment
Assistant Professor, Data Science Major *Yangsan, Korea*

- Research interests: AI for Science, industrial AI and decision-making, evidence-centered AI, and continuous-time modeling.
- Principal Investigator, Scientific Inference Lab.

UCLA Health, David Geffen School of Medicine, UCLA September 2023 to August 2026
Postdoctoral Researcher; Advisor: Prof. Alex Bui *Los Angeles, CA, US*

- Named Nominee for the 2025 UCLA Chancellor's Award for Postdoctoral Research.
- Principal Investigator of the Postdoctoral Fellowship for Overseas Research (Nurturing Next-generation Researchers) from 2024 to 2025, supported by the National Research Foundation of Korea (NRF).
- Recipient of a Fellowship in the Biomedical UniStar Training Program from 2023 to 2024 as a visiting postdoctoral researcher, supported by the Korea Health Industry Development Institute (KHIDI).
- Contributed to the UCLA Depression Grand Challenge, developing trustworthy AI models for clinical and wearable time series with irregular sampling and uncertainty quantification.

Industry Intelligentization Institute, UNIST March 2023 to November 2024
Postdoctoral Researcher; Advisor: Prof. Sungil Kim *Ulsan, Korea*

- Collaborated on advanced AI models for industrial systems with the UNIST Data Analytics Lab, focusing on real-world data applications and model deployment.
- Held a dual affiliation with UNIST and UCLA from September 2023 to November 2024, conducting research at UCLA Health as a Visiting Postdoctoral Researcher supported by institutional funding and government grants.

Data Analytics Lab, Department of Industrial Engineering, UNIST March 2018 to February 2023
Graduate Researcher; Advisor: Prof. Sungil Kim *Ulsan, Korea*

- Founding member of the UNIST Data Analytics Lab, contributing to the establishment of research directions and coordination of industry collaborative projects.
- Led applied research projects on time series modeling and anomaly detection across multiple domains.
- Developed end-to-end machine learning pipelines using real-world sensor, trajectory, and operational data in collaboration with industry partners including port authorities, logistics firms, and energy institutions.

Language Technologies Institute, Carnegie Mellon University (CMU) January 2020 to July 2020
Visiting Student and Researcher; Advisors: Daniel Plakosh and Jeffrey Gennari *Pittsburgh, PA, US*

- Recipient of a scholarship from the Intensive Program in Artificial Intelligence (High-Potential Individuals Global Training Program), supported by the Institute for Information & Communication Technology Planning & Evaluation (IITP), Republic of Korea.
- Completed intensive graduate-level coursework in Natural Language Processing and Applied Machine Learning.
- Led a project team in designing and developing a proof-of-concept conversational AI agent, covering dialogue flow design and end-to-end implementation. [demo]

Global Research and Development (R&D) Center, General Motors (GM) February 2017 to August 2017
Visiting Scientist; Advisors: Dr. Wayne Cai and Dr. Seog-Chan Oh *Warren / Detroit, MI, US*

- Recipient of the UNIST/GM internship fellowship and the FISITA Travel Bursary Scholarship.
- Researcher at the Manufacturing Systems Research Laboratory, analyzing real-world experimental and simulation data to enhance the efficiency and quality of automotive manufacturing processes.
- Conducted analytical analysis using a design of experiments and developed a multi-platform welding simulation tool to optimize process parameters and predict quality outcomes, supporting Digital Twin initiatives. [demo]

Accenture Korea Ltd. October 2016 to January 2017
Research Assistant for the Lotte Duty Free Shop New Generation System Integration Project *Seoul, Korea*

- Supported project management operations (PMO) for a large-scale new generation system integration.

Institut für Theoretische Physik und Astrophysik (ITAP) July 2014 to September 2014
Research Internship; Advisor: Prof. Sebastian Wolf *Kiel, Germany*

- Conducted astrophysics research on the mechanism of interstellar dust alignment for star formation, through the International Association for the Exchange of Students for Technical Experience (IAESTE) program.

Education

UNIST March 2017 to February 2023
Ph.D. in Industrial Engineering *Ulsan, Korea*

- Advisor: Prof. Sungil Kim, [Data Analytics Lab].
- Thesis: "A Comprehensive Study of Deep Learning for Real-World Multivariate Time Series Classification."
- Began in Technology and Innovation Management (2017) and re-enrolled in Industrial Engineering (2018).

UNIST March 2015 to February 2017
Master in Technology and Innovation Management *Ulsan, Korea*

- Advisor: Prof. KwangWook Gang.
- Graduation capstone project: "Discussion of the meaning and components of the fourth industrial revolution, and recommendations for Korea to prepare for the fourth industrial revolution."

UNIST March 2011 to February 2015

B.Sc. in Physics, dual minors in Mechanical Engineering and Advanced Materials Science *Ulsan, Korea*

- Valedictorian (February 2015), Magna cum laude.
- President of Club Union (2013) and President of the Electrical and Computer Engineering Student Council (2012).

Publications

* Co-first author, † Co-corresponding author

Working Papers (In Preparation and Under Review)

- **Oh, Y.**, Cohen, Z., Akre, S., Zbozinek, T., Craske, M. G., & Bui, A. A. T. (in preparation). Predicting depression states using a multi-factor ensemble approach.
- **Oh, Y.**, Rahrooh, A., Talton, L., & Bui, A. A. T. (in preparation). Shifting opinions about generative AI: How an interactive session informs perceptions about its use in research and education.
- **Oh, Y.**, & Bui, A. A. T. (in preparation). Guiding attention with information flow dynamics for time series forecasting.
- **Oh, Y.**, & Bui, A. A. T. (in preparation). Multi-scale differential transformer for long-range forecasting of influenza-like illness.
- **Oh, Y.**, & Yeom, J. (in preparation). Leveraging LLM for sentiment detection in news headlines.
- **Oh, Y.**, Koh, G., Shin, K., Moon, J., & Kim, S. (in preparation). Real-time risk assessment of thyroid function abnormality using irregularly sampled heart rate records.
- **Oh, Y.**, Kwak, J., & Kim, S. (in preparation). Semi-supervised spatio-temporal graph neural networks for detecting non-recurrent traffic congestion in intelligent transportation systems.

Peer-reviewed Conference Proceedings

[C15] **Oh, Y.**, & Bui, A.† (2026). Silent failures in federated personalization of foundation models. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2* (pp. 13199-13206). Association for Computing Machinery. Blue Sky Ideas Track, Jeju Island, Republic of Korea. [paper] [arxiv] [code]

[C14] **Oh, Y.** (2026). Position: State-of-the-art claims require state-of-the-art evidence. In *Proceedings of the 43rd International Conference on Machine Learning* (pp. 171583-171601). PMLR. Position Paper Track, Seoul, Republic of Korea. [paper] [arxiv] [code]

[C13] **Oh, Y.**, Zheng, H. W., Feng, J., & Bui, A. A. T. (2026). Survey-aware machine learning: A guideline for valid population health inference based on scoping review. In *Proceedings of the 7th Conference on Health, Inference, and Learning* (pp. 198-220). PMLR. Seattle, WA, USA. [paper] [arxiv] [code]

[C12] **Oh, Y.**, Lim, D.-Y.†, & Kim, S.† (2026). FlowPath: Learning data-driven manifolds with invertible flows for robust irregularly-sampled time series classification. *Proceedings of the AAAI Conference on Artificial Intelligence, 40*(29), 24594-24603. [paper] [arxiv] [code]

[C11] Lee, J.*, **Oh, Y.***, Kim, S.†, & Lim, D.-Y.† (2026). Continuum dropout for neural differential equations. *Proceedings of the AAAI Conference on Artificial Intelligence, 40*(27), 22796-22804. [paper] [arxiv] [code]

[C10] **Oh, Y.**, Lim, D., Kim, S.†, & Bui, A. A. T.† (2025). TANDEM: Temporal attention-guided neural differential equations for missingness in time series classification. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management* (pp. 2232-2242). Association for Computing Machinery. [paper] [code]

[C09] **Oh, Y.**, Kam, S., Lim, D., & Kim, S.† (2025). Modeling irregular astronomical time series with neural stochastic delay differential equations. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management* (pp. 5068-5073). Association for Computing Machinery. [paper]

[C08] **Oh, Y.**, Kam, S., Lee, J., Lim, D.-Y., Kim, S.†, & Bui, A. A. T.† (2025). Comprehensive review of neural differential equations for time series analysis. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25* (pp. 10621-10631). International Joint Conferences on Artificial Intelligence Organization. [paper] [arxiv] [slide] [poster] [webpage]

[C07] **Oh, Y.**, & Bui, A.† (2025). Multi-view contrastive learning for robust domain adaptation in medical time series analysis. In *Proceedings of the sixth Conference on Health, Inference, and Learning* (pp. 502-526). PMLR. Best Paper Award. [paper] [arxiv] [code] [poster]

[C06] **Oh, Y.**, Kim, S., & Bui, A. A. T. (2025). Deep interaction feature fusion for robust human activity recognition. In *Human Activity Recognition and Anomaly Detection* (pp. 99-116). Springer Nature. [paper] [slide] [workshop]

[C05] **Oh, Y.***, Lim, D.-Y.*, & Kim, S. (2025). DualDynamics: Synergizing implicit and explicit methods for robust irregular time series analysis. *Proceedings of the AAAI Conference on Artificial Intelligence, 39*(18), 19730-19739. [paper] [arxiv] [code] [slide] [poster]

[C04] **Oh, Y.***, Lim, D.*, & Kim, S. (2024). Stable neural stochastic differential equations in analyzing irregular time series data. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. Spotlight Presentation (5%). [paper] [arxiv] [code] [slide] [poster] [news]

[C03] **Oh, Y.** (2022). Multivariate times series classification using multichannel CNN. In L. D. Raedt (Ed.), *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI-22* (pp. 5865-5866). International Joint Conferences on Artificial Intelligence Organization. [paper] [slide]

[C02] **Oh, Y.**, & Kim, S. (2021). Multi-channel convolution neural network for gas mixture classification. In *2021 International Conference on Data Mining Workshops (ICDMW)* (pp. 1094-1095). [paper] [poster]

[C01] **Oh, Y.**, & Kim, S. (2019). Exploiting logistics anomaly detection using maritime big data. *IISE Annual Conference. Proceedings*, 964-969. [paper] [slide]

Journal Papers (Published)

[J13] **Oh, Y.**, Kwak, J., & Kim, S.† (2026). Predicting non-recurrent congestion impact: A pattern-based approach for speed drop ratio prediction using weighted k-nearest neighbors. *Computers & Industrial Engineering, 213*, 111769. [paper]

[J12] Xue, Y., Zhu, Y., Zhuang, L., **Oh, Y.**, Taira, R., Aberle, D. R., Prosper, A. E., Hsu, W., & Lin, Y.† (2025). SmokeBERT: A bidirectional encoder representations from transformers-based model for quantitative smoking history extraction from clinical narratives to improve lung cancer screening. *JCO Clinical Cancer Informatics, 9*, e2500223. [paper] [code]

[J11] **Oh, Y.**, Kim, H., & Kim, S.† (2025). TSSI: Time series as screenshot images for multivariate time series classification using convolutional neural networks. *Computers & Industrial Engineering, 209*, 111393. [paper] [code]

[J10] **Oh, Y.**, Koh, G., Kwak, J., Shin, K., Kim, G.-S., Lee, M. J., Choung, H., Kim, N., Moon, J. H.†, & Kim, S.† (2025). TAOD-Net: Automated detection and analysis of thyroid-associated orbitopathy in facial imagery. *Computers & Industrial Engineering, 203*, 111024. [paper] [code]

[J09] **Oh, Y.**, Yoon, K., Park, J., & Kim, S.† (2025). Comparative evaluation of VAE-based monitoring statistics for real-time anomaly detection in AIS data. *Maritime Policy & Management, 52*(4), 609-626. [paper]

[J08] **Oh, Y.**, & Kim, S.† (2024). Multi-modal lifelog data fusion for improved human activity recognition: A hybrid approach. *Information Fusion, 110*, 102464. [paper] [code]

[J07] **Oh, Y.**, Lim, C., Lee, J., Kim, S., & Kim, S.† (2024). Multichannel convolution neural network for gas mixture classification. *Annals of Operations Research, 339*(1-2), 261-295. [paper] [code]

[J06] **Oh, Y.**, & Kim, S. (2024). Grid-based bayesian bootstrap approach for real-time detection of abnormal vessel behaviors from AIS data in maritime logistics. *IEEE Transactions on Automation Science and Engineering, 21*(4), 6680-6692. [paper] [code] [slide]

[J05] **Oh, Y.**, Lee, J., & Kim, S.† (2023). Sensor drift compensation for gas mixture classification in batch experiments. *Quality and Reliability Engineering International, 39*(6), 2422-2437. [paper]

[J04] **Oh, Y.**, Kwak, J., & Kim, S.† (2023). Time delay estimation of traffic congestion propagation due to accidents based on statistical causality. *Electronic Research Archive, 31*(2), 691-707. [paper] [code]

[J03] Lee, J., Kwak, J., **Oh, Y.**, & Kim, S.† (2023). Quantifying incident impacts and identifying influential features in urban traffic networks. *Transportmetrica B: Transport Dynamics, 11*(1), 279-300. [paper]

[J02] **Oh, Y.**, Kim, H., Lee, D., & Kim, S.† (2021). Simulation-based anomaly detection in nuclear reactors. *Journal of the Korean Institute of Industrial Engineers, 47*(2), 130-143. [paper]

[J01] **Oh, Y.**, Kim, N., & Kim, S.† (2021). Transfer learning based approach for mixture gas classification. *Journal of the Korean Institute of Industrial Engineers, 47*(2), 144-159. [paper]

Conference Presentations

[P18] Lee, Y., Song, J., Ki, J., **Oh, Y.**, Lee, H., Lee, E., & Lee, J. (2026, July). Recovering BRAF signal from individual tumor patches in papillary thyroid carcinoma H&E whole-slide images. In *Medical Imaging with Deep Learning (MIDL 2026)*. Short Papers Poster, Taipei, Taiwan.

[P17] **Oh, Y.**, & Yeom, J. (2026, June). Small language model enhancement strategies in practice: A signal-oriented taxonomy and open questions (extended abstract). *Proceedings of the AAAI Symposium Series, 9*(1), 221-222. Seoul, Republic of Korea. [paper]

[P16] **Oh, Y.** (2026, May). Guiding attention with information flow dynamics for time series forecasting. Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo 2026, Time Series Forecasting & Imputation.

[P15] **Oh, Y.**, Lim, D., Kim, S., & Bui, A. A. T. (2026, January). TANDEM: Temporal attention-guided neural differential equations for missingness in time series classification. Singapore ACM SIGKDD Symposium 2026, Poster Presentation (non-archival; CIKM 2025 paper). Singapore ACM SIGKDD Chapter.

[P14] **Oh, Y.**, Lim, D., & Kim, S. (2025, November). Neural differential equations for continuous-time analysis. The 34th ACM International Conference on Information and Knowledge Management (CIKM) 2025. Association for Computing Machinery (ACM). [paper] [slide] [webpage]

[P13] **Oh, Y.** (2025, October). Continuous-time modeling of complex operational systems with neural differential equations. Institute for Operations Research and the Management Sciences (INFORMS) Annual Meeting 2025.

[P12] **Oh, Y.**, & Bui, A. (2025, June). Multi-scale transformer for long-term time series forecasting. Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo 2025, Data Analytics and Information Systems (DAIS) Best Track Paper Competition. Best Paper Competition Finalist. [paper] [slide]

[P11] Lin, Y., Zhu, Y., **Oh, Y.**, Zhuang, L., Taira, R., Prosper, A.E., & Hsu, W. (2024, November). Extracting quantitative smoking history from clinical narratives using zero-shot guideline-based large language model prompting. IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'24). IEEE Engineering in Medicine and Biology Society (IEEE EMBS). [paper] [poster]

[P10] **Oh, Y.** & Kim, S. (2024, August). Grid-based bayesian bootstrap approach for real-time detection of abnormal vessel behaviors from AIS data in maritime logistics. IEEE 20th International Conference on Automation Science and Engineering. Institute of Electrical and Electronics Engineers (IEEE). [paper] [code] [slide]

[P09] **Oh, Y.**, Lim, D. & Kim, S. (2024, May). Integrating neural controlled differential equations and neural flow for comprehensive irregularly-sampled time series analysis. Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo 2024, Data Analytics and Information Systems (DAIS) Best Track Paper Competition. Best Paper Competition Finalist. [slide]

[P08] **Oh, Y.** & Kim, S. (2023, June). Flow-based neural differential equations for time series analysis. Conference of Korea Computer Congress 2023 (KCC 2023). Korean Institute of Information Scientists and Engineers (KIISE). Best Paper Award. [slide]

[P07] **Oh, Y.** & Kim, S. (2023, June). Heterogeneous model fusion for enhanced sensor-based human activity recognition. Conference of Korea Computer Congress 2023 (KCC 2023). Korean Institute of Information Scientists and Engineers (KIISE). Best Oral Presentation Award. [slide]

[P06] **Oh, Y.** & Kim, S. (2023, May). Neural stochastic differential equations based on the ornstein-uhlenbeck process. Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo 2023, Data Analytics and Information Systems (DAIS) Best Track Paper Competition. Best Paper Competition Finalist. [slide]

[P05] **Oh, Y.** & Kim, S. (2022, December). Lifelog data fusion approach for emotion recognition. Conference of Korea Software Congress 2022 (KSC 2022). Korean Institute of Information Scientists and Engineers (KIISE). [slide]

[P04] **Oh, Y.**, Koh, G., Shin, K., Moon, J. & Kim, S. (2022, November). Real-time risk assessment of thyroid function abnormality using irregularly-sampled heart rate records. The 7th Joint Conference of Korean Artificial Intelligence Association (JKAIA), Fall 2022. Korean Artificial Intelligence Association (KAIA). [poster]

[P03] **Oh, Y.** & Kim, S. (2022, November). Irregularly sampled time series classification using neural stochastic differential equation. The 7th Joint Conference of Korean Artificial Intelligence Association (JKAIA), Fall 2022. Korean Artificial Intelligence Association (KAIA). [poster]

[P02] **Oh, Y.** & Kim, S. (2021, July). Multi-channel convolution neural network for gas mixture classification. The 4th Joint Conference of Korean Artificial Intelligence Association (JKAIA) 2021 / Conference of Korean Artificial Intelligence Association, Summer 2021. Korean Artificial Intelligence Association (KAIA). [paper] [poster]

[P01] **Oh, Y.** & Kim, S. (2021, May). Logistics anomaly detection with maritime big data: A bootstrap approach. Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo 2021. Best Paper Award. [slide]

Peer-reviewed or Invited Workshop Presentations

[W17] **Oh, Y.**, Talton, L., & Bui, A. (2026, August). Engagement intensity as a learner-modeling signal for adaptive AI ethics instruction. Proceedings of the AI for Education Day at SIGKDD 2026, co-located with the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026), CEUR Workshop Proceedings, Vol. 4277. Short Paper, Jeju Island, Republic of Korea. [paper] [arxiv]

[W16] **Oh, Y.**, & Bui, A. (2026, July). Period-aware inductive bias versus scale on influenza-like illness forecasting. Workshop on Forecasting as a New Frontier of Intelligence, The 43rd International Conference on Machine Learning (ICML 2026). International Conference on Machine Learning (ICML). [paper] [workshop]

[W15] **Oh, Y.**, & Bui, A. (2026, July). Context-aware neural SDEs for robust irregular time-series classification. Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM), The 43rd International Conference on Machine Learning (ICML 2026). International Conference on Machine Learning (ICML). [paper] [workshop]

[W14] **Oh, Y.** (2026, July). Uncertainty-aware modeling of continuous-time interacting systems via relational flows. Symposium on Probabilistic Machine Learning (ProbML 2026). Workshop Track. [workshop]

[W13] **Oh, Y.**, & Yeom, J. (2025, December). Confidence-gated LLM synthesis for enhanced multi-class sentiment analysis in financial texts. Workshop on Generative AI in Finance, The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025). Neural Information Processing Systems (NeurIPS). [paper] [workshop]

[W12] **Oh, Y.**, Cohen, Z., Akre, S., Zbozinek, T., Craske, M., & Bui, A. A. T. (2025, October). Interpretable multi-factor framework for longitudinal mental health monitoring. Institute for Operations Research and the Management Sciences (INFORMS) Workshop on Data Science 2025. [workshop]

[W11] **Oh, Y.**, & Bui, A. (2025, August). The compounded risks of federated foundation model personalization under dataset shift. International Workshop on Federated Learning with Generative AI, In Conjunction with IJCAI 2025 (FedGenAI-IJCAI'25). International Joint Conference on Artificial Intelligence (IJCAI). [slide] [workshop]

[W10] **Oh, Y.**, & Bui, A. (2025, March). Integrating transfer entropy into transformer for time series forecasting. AAAI'25 Workshop on Artificial Intelligence with Causal Techniques. Association for the Advancement of Artificial Intelligence (AAAI). [paper] [slide] [poster] [workshop]

[W09] **Oh, Y.**, & Bui, A. (2025, February). Multi-view contrastive learning for medical time series domain adaptation. AI for Medicine and Healthcare, AAAI Bridge Program 2025, The 39th Annual AAAI Conference on Artificial Intelligence. Association for the Advancement of Artificial Intelligence (AAAI). [slide] [workshop]

[W08] **Oh, Y.**, Kam, S., Lim, D. & Kim, S. (2024, October). Advancing irregular time series classification in astronomy with neural stochastic differential equations. Institute for Operations Research and the Management Sciences (INFORMS) Workshop on Data Science 2024. [slide] [workshop]

[W07] **Oh, Y.**, Lim, D., Kim, S. & Bui, A. (2024, August). Attention-guided neural differential equations framework for missingness in time series classification. The 10th Mining and Learning from Time Series Workshop, The ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD) 2024. [poster] [workshop]

[W06] **Oh, Y.**, Kim, S. & Bui, A. (2024, August). Deep interaction feature fusion for robust human activity recognition. 4th International Workshop on Deep Learning for Human Activity Recognition, The 33rd International Joint Conference on Artificial Intelligence (IJCAI) 2024. [paper] [slide] [workshop]

[W05] **Oh, Y.**, Kam, S., Lim, D. & Kim, S. (2024, May). Neural langevin-type stochastic differential equations for astronomical time series classification under irregular observations. AI4DifferentialEquations in Science Workshop, The Twelfth International Conference on Learning Representations (ICLR) 2024. [poster] [workshop]

[W04] **Oh, Y.**, Lim, D. & Kim, S. (2024, February). Invertible solution of neural differential equations for analysis of irregularly-sampled time series. Artificial Intelligence for Time Series (AI4TS) Workshop, The 38th Annual AAAI Conference on Artificial Intelligence. Association for the Advancement of Artificial Intelligence (AAAI). [poster] [workshop]

[W03] **Oh, Y.**, Kam, S., Lim, D. & Kim, S. (2024, February). Enhancing astronomical time series classification with neural stochastic differential equations under irregular observations. Knowledge Guided ML (KGML) Bridge Program, The 38th Annual AAAI Conference on Artificial Intelligence. Association for the Advancement of Artificial Intelligence (AAAI). [slide] [poster] [workshop]

[W02] **Oh, Y.**, Lim, D. & Kim, S. (2023, December). Unifying neural controlled differential equations and neural flow for irregular time series classification. The Symbiosis of Deep Learning and Differential Equations III, Advances in Neural Information Processing Systems. [poster] [workshop]

[W01] **Oh, Y.**, Kwak, J. & Kim, S. (2022, October). Time delay estimation of traffic congestion based on statistical causality. 3rd Workshop on Data-driven Intelligent Transportation (DIT 2022), the 31st ACM International Conference on Information and Knowledge Management (CIKM) 2022. Association for Computing Machinery (ACM). [slide] [workshop]

Grants and Fellowships

Principal Investigator

Postdoctoral Fellowship for Overseas Research (Nurturing Next-generation Researchers)

- Role: Principal Investigator.
- Research Title: "Approach to Detect Distribution Shifts Over Time and Evaluate Model Trustworthy with Retraining in Longitudinal Medical Data."
- Source: National Research Foundation of Korea (NRF).
- Amount: KRW 57,143,000 (approximately USD 41,486).
- Duration: September 1, 2024 to August 31, 2025.

Lead Scientist

NVIDIA Academic Grant Program

- Role: Lead Scientist on the project team.
- Project Title: Neural SDE-Augmented GPU-Accelerated Pharmacokinetic Simulation Framework.
- Principal Investigator: Alex A. T. Bui, UCLA.
- Source: NVIDIA.
- Support: Competitive in-kind cloud computing grant providing a dedicated 8xH100 cloud instance.
- Duration: July 1, 2026 to December 31, 2026.

Researcher & Trainee

Biomedical UniStar Training Program

- Selective program supporting collaborative research between UNIST and the UCLA David Geffen School of Medicine.
- Source: Korea Health Industry Development Institute (KHIDI).
- Amount: approximately USD 50,000 as of March 2023.
- Duration: September 1, 2023 to August 31, 2024.

Intensive Program in Artificial Intelligence (High-Potential Individuals Global Training Program)

- Competitive, fully funded program for intensive training at Carnegie Mellon University (CMU).
- Source: Institute for Information & Communication Technology Planning & Evaluation (IITP).
- Amount: approximately USD 47,500 as of January 2020, covering education fee, stay expenses, and airfare.
- Duration: January 13, 2020 to July 3, 2020.

Honors and Awards

Honors

- **CIRTL Scholar** (2026, March). Center for the Integration of Research, Teaching and Learning (CIRTL), UCLA.
- **Chancellor's Award for Postdoctoral Research Nominee** (2025, November). UCLA.
- **Valedictorian** (2015, February). B.Sc. in Physics, UNIST.

Research Awards

- **Energy Analytics Challenge Finalist** (2026, May). "Transform-First Forecasting for County-Level Power Outages," Energy Systems Division (ESD), Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo.
- **Best Paper Award** (2025, June). "Multi-View Contrastive Learning for Robust Domain Adaptation in Medical Time Series Analysis," Models & Methods Track, The Sixth Conference on Health, Inference, and Learning (CHIL).
- **Best Paper Competition Finalist** (2025, May). "Multi-scale Transformer for Long-term Time Series Forecasting," Data Analytics and Information Systems (DAIS) Division, IISE Annual Conference & Expo. Recognized as a finalist for three consecutive years: 2023, 2024, and 2025.
- **Best Paper Competition Finalist** (2024, May). "Integrating Neural Controlled Differential Equations and Neural Flow for Comprehensive Irregularly-sampled Time Series Analysis," Data Analytics and Information Systems (DAIS) Division, IISE Annual Conference & Expo.
- **Best Paper Award** (2023, June). "Flow-based Neural Differential Equations for Time Series Analysis," Korea Computer Congress (KCC), Korean Institute of Information Scientists and Engineers (KIISE).

- **Best Oral Presentation Award** (2023, June). "Heterogeneous Model Fusion for Enhanced Sensor-Based Human Activity Recognition," AI Division, Korea Computer Congress (KCC), KIISE.
- **Best Paper Competition Finalist** (2023, May). "Neural Stochastic Differential Equations Based on the Ornstein-Uhlenbeck Process," Data Analytics and Information Systems (DAIS) Division, IISE Annual Conference & Expo.
- **Best Paper Award** (2021, May). "Logistics Anomaly Detection with Maritime Big Data: A Bootstrap Approach," Logistics & Supply Chain (LSC) Division, IISE Annual Conference & Expo.
- **Encouragement Award** (2019, December). "Strategies for Building a Startup Ecosystem for Port Smartization," Graduate Student Academic Paper Competition, Ulsan Development Institute.
- **Excellence Award** (2019, November). "Suggestion for Smart Port in Industry 4.0," International Conference on Maritime, Academic Paper Competition, Dongseo University & Ministry of Maritime Affairs and Fisheries.

Academic Scholarship

- **UNIST Graduate Scholarship for Academic Excellence (Ph.D. degree)** (2017 to 2022). UNIST.
- **FISITA Travel Bursary Scholarship** (2017, May). International Federation of Automotive Engineering Societies.
- **UNIST/GM Internship Fellowship** (2016, November). UNIST and General Motors Global R&D Center.
- **UNIST Graduate Scholarship for Academic Excellence (Master degree)** (2015 to 2016). UNIST.
- **UNIST Undergraduate Scholarship for Academic Excellence** (2011 to 2014). UNIST.

Competition

- **Excellence Award** (2023, March). "EMR Data Analysis Project Case Presentation," Invited Talk from Professional Training for Medical Data Analysis, Medical Informatics Homecoming Day, Asan Medical Center & Ministry of Health and Welfare.
- **Encouragement Award** (2021, February). "A Comprehensive Analysis of Regional Commercial Districts and Their Role in Supporting the Success of Young Entrepreneurs," COMPAS Urban Data Analysis Competition, Korea Land & Housing Corporation (LH).
- **1st Place, Case Competition** (2020, November). "Predicting Reoperation in Breast Cancer Using Presurgical Ultrasound Records," EMR Data Analysis Team Project, Professional Training for Medical Data Analysis, Asan Medical Center & Ministry of Health and Welfare.
- **Top Excellence Award (UNIST President's Award)** (2018, December). "Unipillow: AI-based Personalized Smart Pillow for Sleep and Neck Health," Team UNIVENGERS, 2018 Medical Startup Festa, Ulsan Center for Creative Economy and Innovation in partnership with Ulsan University Hospital and University of Ulsan.
- **Joint 1st Place, Business Model Competition** (2017, November). "Smart Headlamp System for Personal Mobility," Minister Award, 2017 Korea Technology Commercialization Fair, Korea Ministry of Trade, Industry and Energy (MOTIE) & Korea Institute for Advancement of Technology (KIAT).

Patents

Co-inventor across all listed filings.

Apparatus and method for human activity recognition through hybrid fusion of dynamic and static data

- KR 10-2024-0033382 (applied March 8, 2024; application pending).

Encoding apparatus and method for converting time series data into images

- KR 10-2754134 (granted January 8, 2025); applied as KR 10-2022-0183341 (December 23, 2022).
- US 12,725,326 B2 (granted September 1, 2026; applied as US 18/395,620 on December 24, 2023).

Method, computer device, and computer program to predict road congestion propagation using pattern matching

- KR 10-2608343 (granted November 27, 2023); applied as KR 10-2021-0120229 (September 9, 2021).
- JP 7453296 (granted March 11, 2024); applied as JP 2022-137508 (August 31, 2022; pub. no. JP2023039925A).
- Joint applicants: Naver Corp. and UNIST.

Method, computer device, and computer program to predict propagation delay time of road congestion using transfer entropy

- KR 10-2604575 (granted November 16, 2023); applied as KR 10-2021-0119772 (September 8, 2021).
- JP 7453297 (granted March 11, 2024); applied as JP 2022-137509 (August 31, 2022; pub. no. JP2023039418A).
- Joint applicants: Naver Corp. and UNIST.

Method of anomaly detection of vessels applying Bayesian bootstrap

- KR 10-2534357 (granted May 16, 2023); applied as KR 10-2020-0178651 (December 18, 2020).

Method and device for compensating for sensor drift

- KR 10-2364019 (granted February 14, 2022); applied as KR 10-2020-0101037 (August 12, 2020).
- PCT/KR2021/010729 (applied August 12, 2021; prosecution discontinued); US 18/012,626 (applied December 22, 2022; prosecution discontinued).

Method and apparatus for determining delay possibility of shipment

- KR 10-2250354 (granted May 4, 2021); applied as KR 10-2019-0158913 (December 3, 2019).
- PCT/KR2020/015854 (applied November 12, 2020; prosecution discontinued).

Academic Services

Professional Development

- **Implementing and Communicating a Teaching-as-Research**, CIRTL, UCLA, Spring 2025.
- **Preparing for Teaching-as-Research**, CIRTL, UCLA, Winter 2025.
- **Introduction to Evidence-Based Undergraduate Teaching**, CIRTL, UCLA, Fall 2024.
- **NIH K-award Grant Writing Bootcamp**, UCLA Clinical and Translational Science Institute (CTSI) & DGSOM Office of Physician-Scientist Career Development (OPSCD), UCLA, Summer 2024.
- **Teaching Institute**, The Johns Hopkins University, Summer 2024.

Teaching Experience

- **BX3600087/DS3600087/AB3600087, Machine Learning**, Instructor, Pusan National University, Fall 2026.
- **DS3600082, Data Structure**, Instructor, Pusan National University, Fall 2026.
- **UCLA Teaching & Learning Center Catalyst Grant**, "AI Training Program: Integrating LLMs into Health Professions Education," Workshop Design Team Member, contributing to the design of faculty development workshops on ethical AI integration for medical and nursing educators, 2025 to 2026.
- **UNIST Undergraduate Interdisciplinary Research Project**, advised the student-led project and presentation "Yeo, J., Seo, H., and Shim, S., A Methodology for Predicting Traffic Accidents Using Road Network Structure," KIIE Fall Conference, November 2021. Awarded in the undergraduate industrial engineering project contest, UNIST, Spring to Fall 2021.
- **MGT205, Financial Accounting**, TA, UNIST, Fall 2019.
- **MGT206, Managerial Accounting**, TA, UNIST, Fall 2019.
- **MGT206, Managerial Accounting**, TA, UNIST, Fall 2018.
- **MGE362, Quality Management**, TA, UNIST, Spring 2018.
- **TIM614, Integration of IT, Manufacturing, and Operational Systems**, TA, UNIST, Fall 2017.
- **MGT473, Entrepreneurship and Venture Management**, TA, UNIST, Spring 2016.
- **AHS230, Recent Trends in Science and Technology Business of Korea**, TA, UNIST, Fall 2015.
- **MGT410, Psychology and Marketing**, TA, UNIST, Spring 2015.

Organizer & Presenter

- **Tutorial**: **Oh, Y.**, Lim, D., & Kim, S. Neural Differential Equations for Continuous-Time Analysis. The 34th ACM International Conference on Information and Knowledge Management (CIKM 2025), November 2025. Tutorial track.

Reviewer / Program Committee

- **Journal**: Information Fusion, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Automation Science and Engineering (T-ASE), Expert Systems with Applications, Engineering Applications of Artificial Intelligence (EAAI), Applied Soft Computing, Neurocomputing, Computers and Industrial Engineering (CAIE), Computers in Biology and Medicine (CIBM), Behaviour Research and Therapy, Accident Analysis and Prevention, Transportation Research.
- **Conference**: International Conference on Learning Representations (ICLR), AAAI Conference on Artificial Intelligence (AAAI), Neural Information Processing Systems (NeurIPS), ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), Conference on Health, Inference, and Learning (CHIL), Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo.

Research Projects

UCLA Depression Grand Challenge September 2023 to Present
AI-based solutions to advance depression research and treatment strategies *UCLA*

- UCLA-led research initiative with the stated goal of cutting the global burden of depression in half by 2050.
- Collaborate with Apple and UCLA Health on research using mobile technology and wearable devices to identify early indicators of depression and track treatment effectiveness.
- Develop and implement AI-based solutions for studies including DMHS (Digital Mental Health Study) and OPTIMA (Operationalizing Digital PhenoTyping In the Measurement of Anhedonia), analyzing digital biomarkers to measure depression symptoms.

Human-Centered Carbon Neutral Global Supply Chain Research Center January 2023 to August 2023
Proposal writing support for a multi-disciplinary research center grant *UNIST*

- Contributed to the core research proposal for a large-scale multi-disciplinary research center focused on carbon neutrality and sustainable supply chain innovation: Safe and Clean Supply Chain (SCSC) center at Pusan National University.
- Co-designed the data-driven analytics framework for the Data Science Group.

THYROSCOPE Inc. August 2021 to August 2023
Classification for eye disease related to thyroid illness using computer vision models *UNIST*

- Developed a classification model for eye disease related to thyroid illness, using pictures taken by a DSLR camera.
- Engineered a visual symptom recognition model using a vision transformer (TAOD-Net).
- Designed a domain adaptation method for image data from different smartphone devices.

THYROSCOPE Inc. August 2021 to August 2023
Thyroid hormone prediction with resting heart rate from wearable device *UNIST*

- Developed a prediction model for thyroid hormone using multi-modal sensor data from a wearable device.
- Designed a predictive framework for determining appropriate thyroid drug dosage from wearable sensor data.

NAVER Maps November 2020 to October 2022
Traffic prediction and congestion modeling using GPS trajectory data *UNIST*

- Developed a traffic prediction model using historical and real-time data.
- Developed and validated algorithms for predicting traffic congestion from quantitative and qualitative perspectives.
- Conducted causal relationship analysis using traffic incident-oriented speed data.

Korea Hydro & Nuclear Power Co., Ltd. September 2020 to May 2021
ML-based diagnosis and status analysis for nuclear reactor cores *UNIST*

- Developed a recognition model to diagnose and analyze the status of a reactor core in a nuclear power plant.

CyberLogitec Co., Ltd. June 2019 to December 2019
Development of optimization algorithm and solver for large-scale optimization problems *UNIST*

- Developed a pipeline for large-scale optimization problems using open source solvers including CBC.

Ulsan Port Authority January 2019 to December 2019
Development of smart port and logistics system *UNIST*

- Analyzed Automatic Identification System (AIS) data for maritime surveillance in a smart port system.

Taesung Environmental Research Institute Co. August 2018 to July 2020
Analysis of sensor data from gas mixtures in a manufacturing environment *UNIST*

- Constructed research setup and developed pipeline including database setup to deployment.
- Developed linear and non-linear models for multi-class classification using sensor data.

Consulting Experiences

UNIST Strategic Consulting Association (UNISCA) 2017 to 2021
Co-founder (2017), Senior Leadership Team (2017-2019), Senior Member (2019-2021) *UNIST*

- Co-founded and led UNISCA, the strategic consulting association at UNIST, focusing on building successful companies in an innovative and changing era, in collaboration with the Business Intelligence Group (BIG) at the University of Illinois Urbana-Champaign (UIUC).
- Projects: CSL Behring, US (2021), Class 101, KR (2018-2019), IBM, US (2018), Deere & Company, US (2017).

International Consulting Network (ICON) 2016 to 2019
President of UNIST chapter (2016), Senior Leadership Team (2017-2019) *UNIST*

- ICON is an international student consulting organization with chapter partnerships across the United States, South Korea, China, Brazil, Mexico, Switzerland, and Sweden, focused on innovation-strategy projects with a global perspective.
- Projects: Lalubema Sistemas, Brazil (2018), 3D WOOW, China (2017), Greenely, Sweden (2017).

Innovation Immersion Program (IIP) 2015 to 2016
President of UNIST chapter (2015-2016) *UNIST*

- IIP is a worldwide consulting student body for project-based research, hands-on business experiences, and consulting with global clients. IIP has chapters in the United States, South Korea, Sweden, Israel, Singapore, and Taiwan.
- Projects: Daewoong Pharmaceutical Co., Ltd., & BIO-AGE, KR (2016), Agrible, Inc., US (2015).